

FILTRATIONS AND CHOW STABILITY REVISITED

KEWEI ZHANG

ABSTRACT. In this article we express the Chow weight using filtrations, which in turn is related to the slope at infinity of the quantized K-energy along Bergman geodesic rays.

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1. INTRODUCTION

This article is motivated by the problem of finding constant scalar curvature Kähler metrics on polarized Kähler manifolds. The existence of such metrics is conjectured to be equivalent to certain algebro-geometric notion called *K-stability*, which goes back to Tian [26] and Donaldson [11]. For a comprehensive discussion of this subject, we refer the reader to Boucksom–Hisamoto–Jonsson [5]. In this article we will revisit a closely related and well-understood notion, i.e., *Chow stability*, which can be viewed as a “quantization” of K-stability. One aim of this article is to reformulate Chow stability using filtrations and it turns out that such consideration naturally relates Chow stability to the slope stability of certain quantized K-energy. Even though we believe that these contents are well-known to experts (see e.g., Székelyhidi [25] for the same kind of treatment), we decide to revisit the details from a relatively more modern viewpoint, which is also motivated by the author’s recent work [31] on the quantization proof of the Yau–Tian–Donaldson conjecture for twisted Kähler–Einstein metrics. Anyhow, this treatment indeed has some consequences that might look a bit surprising at first glance; see for instance Corollary 1.11, which seems to be new. Another goal of this article is to study the relation between Chow stability and K-stability. This is of course an old topic. But as we shall see, using filtrations, one can extract some useful information that might serve as a new perspective to attack some open problems in this field. Some new stability thresholds characterizing Chow stability and K-stability are also introduced in the end.

1.1. Conventions. Let (X, L) be a polarized Kähler manifold of dimension n , where L is an ample line bundle. For simplicity we will further assume that mL is very ample and

$$H^i(X, mL) = 0 \text{ for all } i \geq 1$$

whenever $m \geq 1$. Set

$$R := R(X, L) := \bigoplus_{m \in \mathbb{N}} R_m,$$

where $R_m := H^0(X, mL)$ and $\mathbb{N} = \{0, 1, 2, \dots\}$. Also put

$$d_m := \dim R_m.$$

After replacing L by a sufficiently divisible multiple we may (and will) assume that the subring

$$R^{(r)} := R(X, rL)$$

is generated in degree 1 for any $r \in \mathbb{N}_{>0}$. These harmless assumptions will make life a lot easier.

1.2. Filtrations and test configurations. We recall some standard facts about filtrations and test configurations. For more details, we refer to [29, 25] and [5, §2].

Definition 1.1. (*Filtration*) We call $\mathcal{F}$ a filtration of R if for any $\lambda \in \mathbb{R}$ and $m \in \mathbb{N}$, there is a subspace $\mathcal{F}^\lambda R_m$ of R_m satisfying

- (1) $\mathcal{F}^{\lambda'} R_m \supseteq \mathcal{F}^\lambda R_m$ for any $\lambda' \leq \lambda$;
- (2) $\mathcal{F}^\lambda R_m = \bigcap_{\lambda' < \lambda} \mathcal{F}^{\lambda'} R_m$;
- (3) $\mathcal{F}^{\lambda_1} R_{m_1} \cdot \mathcal{F}^{\lambda_2} R_{m_2} \subseteq \mathcal{F}^{\lambda_1 + \lambda_2} R_{m_1 + m_2}$ for any $\lambda_1, \lambda_2 \in \mathbb{R}$ and $m_1, m_2 \in \mathbb{N}$;
- (4) $\mathcal{F}^\lambda R_m = R_m$ for $\lambda \leq 0$ and there exists $C > 0$ such that $\mathcal{F}^{Cm} R_m = \{0\}$ for any $m \in \mathbb{N}_{>0}$.

Given a filtration $\mathcal{F}$ of R , let

$$a_{m,i} := \inf\{\lambda \in \mathbb{R}_{\geq 0} \mid \text{codim} \mathcal{F}^\lambda R_m \geq i\}$$

denote the *jumping numbers* of $\mathcal{F}$. We call $\mathcal{F}$ an $\mathbb{N}$ -filtration if all $a_{m,i}$'s are nonnegative integers. Set

$$T_m(\mathcal{F}) := \frac{a_{m,d_m}}{m}, \quad S_m(\mathcal{F}) := \frac{1}{md_m} \sum_{i=1}^{d_m} a_{m,i} \quad \text{and} \quad J_m(\mathcal{F}) := T_m(\mathcal{F}) - S_m(\mathcal{F}).$$

Put

$$T(\mathcal{F}) := \lim_{m \rightarrow \infty} T_m(\mathcal{F}), \quad S(\mathcal{F}) := \lim_{m \rightarrow \infty} S_m(\mathcal{F}) \quad \text{and} \quad J(\mathcal{F}) := \lim_{m \rightarrow \infty} J_m(\mathcal{F}).$$

(The existence of these limits is proved in [3].) One should think of $J(\mathcal{F})$ as certain norm of $\mathcal{F}$. We call $\mathcal{F}$ *trivial* at level m if $J_m(\mathcal{F}) = 0$. Linearly bounded filtrations with possibly negative jumping numbers can be reduced to our nonnegative convention by a degreewise translation: replacing $\mathcal{F}^\lambda R_m$ by $\mathcal{F}^{\lambda - cm} R_m$ adds cm to all jumping numbers in degree m . Both Chow_m and J_m are unchanged by this translation. For $\mathbb{Z}$ -filtrations one may choose c integral.

Definition 1.2 ([11]). A test configuration $(\mathcal{X}, \mathcal{L})$ for (X, L) with exponent r consists of the following data:

- (1) a flat proper morphism π from a variety $\mathcal{X}$ to $\mathbb{C}$;
- (2) a $\mathbb{C}^*$ -action on $\mathcal{X}$ lifting the canonical action on $\mathbb{C}$;
- (3) a $\mathbb{C}^*$ -equivariant (relatively) very ample line bundle $\mathcal{L}$ on $\mathcal{X}$ such that $(\mathcal{X}_1, \mathcal{L}_1) \cong (X, rL)$.

Definition 1.3. An $\mathbb{N}$ -filtration $\mathcal{F}$ of R is called *finitely generated* if the graded $\mathbb{C}[t]$ -algebra (also called the Rees algebra)

$$\bigoplus_{m \in \mathbb{N}} \left(\bigoplus_{\lambda \in \mathbb{Z}} t^{-\lambda} \mathcal{F}^\lambda R_m \right)$$

is *finitely generated*.

Given a finitely generated $\mathbb{N}$ -filtration $\mathcal{F}$ of R , we may find a sufficiently divisible integer $r > 0$ such that

$$\bigoplus_{m \in \mathbb{N}} \left(\bigoplus_{\lambda \in \mathbb{Z}} t^{-\lambda} \mathcal{F}^\lambda R_{mr} \right)$$

is generated in degree 1. Put

$$\mathcal{X} := \text{Proj}_{\mathbb{C}[t]} \bigoplus_{m \in \mathbb{N}} \left(\bigoplus_{\lambda \in \mathbb{Z}} t^{-\lambda} \mathcal{F}^\lambda R_{mr} \right) \quad \text{and} \quad \mathcal{L} := \mathcal{O}_{\mathcal{X}}(1).$$

Then $(\mathcal{X}, \mathcal{L})$ gives rise to a test configuration of (X, L) with exponent r . Equivalently, $(\mathcal{X}, \mathcal{L})$ can be constructed explicitly as follows: Choose a basis $\{s_i\}_{1 \leq i \leq d_r}$ of R_r such that

$$s_i \in \mathcal{F}^{a_{r,i}} R_r \quad \text{for each } 1 \leq i \leq d_r.$$

Then $\{s_i\}$ together with the ‘weights’ $\{a_{r,i}\}$ (i.e., the jumping numbers of $\mathcal{F}$ on R_r) induces a 1-parameter subgroup of $GL(d_r, \mathbb{C})$, which produces the flat family $\mathcal{X}$ inside $\mathbb{P}^{d_r-1} \times \mathbb{C}$ and $\mathcal{L}$ is the pullback of $\mathcal{O}_{\mathbb{P}^{d_r-1} \times \mathbb{C}}(1)$.

Given any integer $r > 0$, a basis $\{s_i\}_{1 \leq i \leq d_r}$ of R_r and d_r non-negative integers $0 \leq \lambda_1 \leq \dots \leq \lambda_{d_r}$, one can construct a finitely generated $\mathbb{N}$ -filtration $\mathcal{F}$ of R as follows: Put for $\lambda \in \mathbb{R}$

$$\mathcal{F}^\lambda R_r := \text{Span}_{\mathbb{C}}\{s_i \mid \lambda_i \geq \lambda\};$$

then one can extend this filtration of R_r to a finitely generated $\mathbb{N}$ -filtration $\mathcal{F}$ of R using [4, Definition 3.18] (see also [25, §3.2]). A filtration thus obtained gives rise to a test configuration $(\mathcal{X}, \mathcal{L})$ with exponent

r , which clearly coincides with the one obtained by considering the 1-parameter subgroup of $GL(d_r, \mathbb{C})$ induced by the $\{s_i\}$ and the weights $\{\lambda_i\}$.

Conversely, any test configuration of (X, L) arises essentially in this way (see [22, Proposition 3.7]). More precisely, given a test configuration $(\mathcal{X}, \mathcal{L})$ of (X, L) with exponent r , then it is induced by a basis $\{s_i\}_{1 \leq i \leq d_r}$ of R_r together with some weights $\{\lambda_i\}_{1 \leq i \leq d_r}$. We may modify (this does not affect the stability notions to be introduced) the $\mathbb{C}^*$ -action on $(\mathcal{X}, \mathcal{L})$ by a constant weight to make $\min_i \{\lambda_i\} = 0$. So each λ_i is a non-negative integer. Then the construction in the previous paragraph gives a finitely generated $\mathbb{N}$ -filtration of R , which we will denote by $\mathcal{F}_{(\mathcal{X}, \mathcal{L})}$. We also put

$$J(\mathcal{X}, \mathcal{L}) := J(\mathcal{F}_{(\mathcal{X}, \mathcal{L})}).$$

This coincides with the J^{NA} -functional of test configurations in the literature.

Definition 1.4 ([5]). *We call a test configuration $(\mathcal{X}, \mathcal{L})$ almost trivial if $J(\mathcal{X}, \mathcal{L}) = 0$. We call $(\mathcal{X}, \mathcal{L})$ trivial if it is a trivial product of (X, L) with $\mathbb{C}$.*

Geometrically, almost trivial means that, after normalization, $(\mathcal{X}, \mathcal{L})$ is a trivial product.

1.3. Chow stability and K-stability. Next we attach a new invariant to filtrations. As we shall see this invariant has fruitful meanings when it comes to the stability notions of (X, L) .

Definition 1.5. *Let $\mathcal{F}$ be a filtration of R . For $m \in \mathbb{N}_{>0}$, we call*

$$\text{Chow}_m(\mathcal{F}) := S(\mathcal{F}) - S_m(\mathcal{F})$$

the Chow invariant of $\mathcal{F}$ at level m .

Definition 1.6. *(Chow stability) We call the pair (X, L) Chow stable at level m if for any test configuration $(\mathcal{X}, \mathcal{L})$ with exponent m , one has*

$$\text{Chow}_m(\mathcal{F}_{(\mathcal{X}, \mathcal{L})}) \geq 0,$$

and the equality holds only when $(\mathcal{X}, \mathcal{L})$ is trivial.

In the view of [29, 25], this definition coincides with the classical one in the literature.

Next, we recall the notion of K-stability, which was defined using the *generalized Futaki invariant*. This invariant first appeared in the work of Ding–Tian [9] and was later reformulated more algebraically by Donaldson [11].

Definition 1.7. *We call the pair (X, L) K-stable if any test configuration $(\mathcal{X}, \mathcal{L})$ with certain exponent r , the generalized Futaki invariant*

$$\text{Fut}(\mathcal{X}, \mathcal{L}) := \lim_{m \rightarrow \infty} 2mr \cdot \text{Chow}_{mr}(\mathcal{F}_{(\mathcal{X}, \mathcal{L})})$$

is always non-negative and it is zero only when $(\mathcal{X}, \mathcal{L})$ is almost trivial. We call the pair (X, L) uniformly K-stable if there exists $\varepsilon > 0$ such that

$$\text{Fut}(\mathcal{X}, \mathcal{L}) \geq \varepsilon J(\mathcal{X}, \mathcal{L})$$

for any test configuration $(\mathcal{X}, \mathcal{L})$ of (X, L) .

1.4. Main results.

Theorem 1.8. *The pair (X, L) is Chow stable at level m if and only if there exists $\varepsilon > 0$ such that*

$$\text{Chow}_m(\mathcal{F}) \geq \varepsilon J_m(\mathcal{F})$$

for any filtration $\mathcal{F}$ of R .

Corollary 1.9. *Assume that (X, L) has discrete automorphism group, and that there exists a cscK metric in $c_1(L)$, then for any filtration $\mathcal{F}$ of R with $J(\mathcal{F}) > 0$ it holds that*

$$S(\mathcal{F}) > S_m(\mathcal{F}) \text{ for all } m \gg 1.$$

Proof. By Donaldson [10], for any $m \gg 1$, (X, L) is Chow stable at level m . Note also that $J_m(\mathcal{F}) > 0$ for $m \gg 1$. So the assertion follows from Theorem 1.8. $\square$

Corollary 1.10. *If (X, L) is Chow stable at level m then $\delta_m(L) > \delta(L)$.*

Recall here that

$$\delta_m(L) := \inf_E \frac{A_X(E)}{S_m(E)} \text{ and } \delta(L) := \inf_E \frac{A_X(E)}{S(E)},$$

where E runs through all the prime divisors over X . Note that $A_X(E)$ is the log discrepancy of E and each E induces a filtration of R so $S(E)$ and $S_m(E)$ have obvious meanings.

Proof. Let $\varepsilon > 0$ be given by Theorem 1.8. Since $|mL|$ is base point free, the filtration induced by any prime divisor E has at least one zero jumping number at level m . Thus

$$S_m(E) \leq \frac{d_m - 1}{d_m} T_m(E), \quad J_m(E) \geq \frac{S_m(E)}{d_m - 1}.$$

Consequently,

$$S(E) \geq \left(1 + \frac{\varepsilon}{d_m - 1}\right) S_m(E).$$

For every E with $S_m(E) > 0$ this gives

$$\frac{A_X(E)}{S_m(E)} \geq \left(1 + \frac{\varepsilon}{d_m - 1}\right) \frac{A_X(E)}{S(E)}.$$

Taking the infimum, with the convention $A_X(E)/S_m(E) = +\infty$ when $S_m(E) = 0$, we obtain

$$\delta_m(L) \geq \left(1 + \frac{\varepsilon}{d_m - 1}\right) \delta(L) > \delta(L),$$

where we have used $\delta(L) > 0$ for an ample polarization. $\square$

The next result might look a bit surprising at first glance. It would be interesting if one can find a purely algebraic proof of this.

Corollary 1.11. *Let X be a K -stable Fano manifold, then for all sufficiently large and divisible $m > 0$ one has*

$$\delta_m(-K_X) > \delta(-K_X).$$

Proof. By the Yau–Tian–Donaldson theorem [7, 27], there exists a Kähler–Einstein metric in $c_1(-K_X)$. Also note that X has discrete automorphism group. Then Donaldson’s work [10] implies that $(X, -K_X)$ is Chow stable at sufficiently high levels. Choose a sufficiently divisible $r > 0$ such that the assumptions in §1.1 holds for $(X, -rK_X)$. Then we can apply the previous corollary to conclude. $\square$

2. THE PROOF

This section is devoted to the proof of Theorem 1.8. To start with, we recall the following well-known analytic characterization of Chow stability; see e.g., [18, 19, 20, 28] for several different proofs.

Theorem 2.1. *The pair (X, mL) is Chow polystable if and only if it admits a balanced embedding into $\mathbb{P}^{d_m-1}$. It is Chow stable if and only if, in addition, its polarized automorphism group $\text{Aut}(X, mL)$ is finite.*

Here Chow polystability means that the orbit of a nonzero lift of the Chow point under $SL(d_m, \mathbb{C})$ is closed; Chow stability also requires a finite projective stabilizer. The polarized automorphism group consists of automorphisms of X preserving the isomorphism class of mL , without including scalar automorphisms of the line bundle. The distinction is necessary when X has continuous polarized automorphisms; see [28, Corollary 27].

We will not recall the definition of balanced embedding as it is not needed in what follows. But instead, one key point we shall recall is the following: As illustrated by Donaldson [12], the existence of balanced embeddings can be detected by certain functionals defined on the m -th Bergman space, which we now describe. Fix a positively curved smooth Hermitian metric h on L with $\omega := -dd^c \log h > 0$ as its curvature form in $c_1(L)$. Put

$$\mathcal{H}_\omega := \left\{ \phi \in C^\infty(X, \mathbb{R}) \mid \omega_\phi := \omega + dd^c \phi > 0 \right\}.$$

Let

$$E(\phi) := \frac{1}{(n+1)V} \sum_{i=0}^n \int_X \phi \omega^i \wedge \omega_\phi^{n-i}$$

denote the Monge–Amère energy for $\phi \in \mathcal{H}_\omega$, where $V := \int_X \omega^n$. We also use its Bedford–Taylor extension to bounded ω -plurisubharmonic potentials. The m -th Bergman space $\mathcal{B}_m$ is a subset in $\mathcal{H}_\omega$, which is defined as

$$\mathcal{B}_m := \left\{ \phi = \frac{1}{m} \log \sum_{i=1}^{d_m} |s_i|_{h^m}^2 \mid \{s_i\} \text{ is a basis of } R_m \right\}.$$

Note that

$$H_m := \int_X h^m(\cdot, \cdot) \omega^n$$

defines a Hermitian inner product on the vector space R_m . Then simple linear algebra shows that, for any $\phi \in \mathcal{B}_m$, there exist an H_m -orthonormal basis $\{s_i\}_{1 \leq i \leq d_m}$ of R_m and a set of real numbers $\{\lambda_i\}_{1 \leq i \leq d_m}$ such that

$$\phi = \frac{1}{m} \log \sum_{i=1}^{d_m} e^{\lambda_i} |s_i|_{h^m}^2.$$

We may thus define the quantized Monge–Ampère energy as

$$E_m(\phi) := \frac{1}{md_m} \sum_{i=1}^{d_m} \lambda_i$$

and the quantized supremum as

$$T_m(\phi) := \frac{\max_{1 \leq i \leq d_m} \{\lambda_i\}}{m}.$$

for $\phi \in \mathcal{B}_m$. The moment-map description in [12, 28], together with the finite-dimensional Kempf–Ness criterion, gives the following analytic criterion for Chow stability. Coercivity here is on the full Bergman space, rather than modulo the action of the polarized automorphism group.

Theorem 2.2. *(X, mL) admits a balanced embedding into $\mathbb{P}^{d_m-1}$ and $\text{Aut}(X, mL)$ is finite if and only if the functional $E - E_m$ is coercive on $\mathcal{B}_m$, namely, there exist $\varepsilon > 0$ and $C > 0$ such that*

$$E(\phi) - E_m(\phi) \geq \varepsilon(T_m(\phi) - E_m(\phi)) - C \text{ for all } \phi \in \mathcal{B}_m.$$

In this case the balanced embedding corresponds to the unique critical point of $E - E_m$ modulo addition of constants.

The Bergman kernel asymptotic expansion in [12] relates $2m(E - E_m)$ to the classical K-energy of Mabuchi, after fixing additive constants. More precisely, for $\phi \in \mathcal{H}_\omega$, let $\phi_m(\phi)$ be the Fubini–Study potential of the inner product

$$\text{Hilb}_m(\phi) := \int_X h^m e^{-m\phi}(\cdot, \cdot) \omega_\phi^n$$

on R_m . Then

$$K_m(\phi_m(\phi)) - K_m(\phi_m(0)) \longrightarrow K(\phi) - K(0).$$

The unnormalized values need not converge. We call

$$K_m(\phi) := 2m(E(\phi) - E_m(\phi)) \text{ for } \phi \in \mathcal{B}_m$$

the *quantized K-energy at level m* .

The following result is clear.

Corollary 2.3. *The pair (X, L) is Chow stable at level m if and only if K_m is coercive on $\mathcal{B}_m$.*

Next we explain how do filtrations of R relate to the above variational picture. Given a filtration $\mathcal{F}$ of R , one can associate a ‘geodesic ray’ $\phi_t^\mathcal{F}$ to $\mathcal{F}$ using the construction in [23]. More precisely, choose an H_m -orthonormal basis $\{s_i\}$ of R_m such that

$$s_i \in \mathcal{F}^{a_{m,i}} R_m \text{ for all } 1 \leq i \leq d_m.$$

Namely $\{s_i\}$ is an H_m -orthonormal basis that is compatible with $\mathcal{F}$. Set for any $t \geq 0$

$$\phi_{m,t}^\mathcal{F} := \frac{1}{m} \log \sum_{i=1}^{d_m} e^{a_{m,i}t} |s_i|_{h^m}^2.$$

Then by definition,

$$S_m(\mathcal{F}) = \frac{E_m(\phi_{m,t}^\mathcal{F})}{t} \text{ and } T_m(\mathcal{F}) = \frac{T_m(\phi_{m,t}^\mathcal{F})}{t} \text{ for all } t > 0.$$

Now for $t \geq 0$ put

$$\phi_t^{\mathcal{F}} := \lim_{m \rightarrow \infty} \left(\sup_{k \geq m} \phi_{k,t}^{\mathcal{F}} \right)^*,$$

where $*$ denotes the upper semi-continuity regularization. By [30, (6.5)] one has

$$S(\mathcal{F}) = \frac{E(\phi_t^{\mathcal{F}})}{t} \text{ for all } t > 0.$$

As a consequence, we obtain that

$$(2.1) \quad \text{Chow}_m(\mathcal{F}) = S(\mathcal{F}) - S_m(\mathcal{F}) = \frac{E(\phi_t^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})}{t} \text{ for all } t > 0.$$

Lemma 2.4. *For any $\mathbb{N}$ -filtration $\mathcal{F}$ of R , one has*

$$\text{Chow}_m(\mathcal{F}) \geq \lim_{t \rightarrow \infty} \frac{E(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})}{t}.$$

Proof. Denote by $\mathcal{F}_m$ the finitely generated $\mathbb{N}$ -filtration of R generated by the filtration $\mathcal{F}|_{R_m}$ of R_m (see [4, Definition 3.18]). So in particular,

$$S_m(\mathcal{F}) = S_m(\mathcal{F}_m) \text{ and } \phi_{m,t}^{\mathcal{F}} = \phi_{m,t}^{\mathcal{F}_m}.$$

On the other hand, for any $k \in \mathbb{N}_{>0}$ one has

$$S_{mk}(\mathcal{F}) \geq S_{mk}(\mathcal{F}_m)$$

So sending $k \rightarrow \infty$ we conclude that

$$S(\mathcal{F}) \geq S(\mathcal{F}_m).$$

Now let $(\mathcal{X}, \mathcal{L})$ be the test configuration of (X, L) induced by $\mathcal{F}_m$ and consider the associated geodesic ray $\phi_t^{\mathcal{F}_m}$. The filtered Bergman limit agrees with the Phong–Sturm ray by [23, Theorem 9.2]. Applying [21, Lemma 4 and Theorem 2] to the polarization mL and the metric h^m , and dividing the potentials by m , gives

$$\sup_{t \geq 0} \|\phi_t^{\mathcal{F}_m} - \phi_{m,t}^{\mathcal{F}_m}\|_{L^\infty(X)} \leq C_{m,\mathcal{F},h}.$$

Indeed, in the disk coordinate w with $t = -\log |w|^2$, the difference between the limiting potential and the potential from the fixed degree-one embedding extends as a bounded function to an equivariant resolution of the total space. This result does not require the total space to be normal or the central fiber to be reduced. The volume normalization in the L^2 inner product changes the fixed-degree potential by a constant, and the terms of order $\log k/k$ in the Bergman approximants vanish in the limit. The constant above is independent of t ; no uniformity in m is required. One may alternatively use [1, Proposition 2.7] after normalization of the total space and identification of its envelope ray.

Since the Monge–Ampère energy satisfies

$$|E(\phi) - E(\psi)| \leq \|\phi - \psi\|_{L^\infty(X)},$$

the same bound applies to the difference of the energies. Therefore

$$S(\mathcal{F}) - S_m(\mathcal{F}) \geq S(\mathcal{F}_m) - S_m(\mathcal{F}_m) = \lim_{t \rightarrow \infty} \frac{E(\phi_t^{\mathcal{F}_m}) - E_m(\phi_{m,t}^{\mathcal{F}_m})}{t} = \lim_{t \rightarrow \infty} \frac{E(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})}{t},$$

as desired. $\square$

Remark 2.5. It is clear from the above proof that the equality

$$(2.2) \quad \text{Chow}_m(\mathcal{F}) = \lim_{t \rightarrow \infty} \frac{E(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})}{t}$$

holds when $\mathcal{F} = \mathcal{F}_{(\mathcal{X}, \mathcal{L})}$ is induced by a test configuration of (X, L) with exponent m . This is probably well-known and is a quantized version of Chi Li's result [15, Theorem 1.7.2].

We are now ready to prove our main result.

Proof of Theorem 1.8. The ‘if’ part follows immediately from the definition.

For the ‘only if’ part, we first treat the case where $\mathcal{F}$ is an $\mathbb{N}$ -filtration of R . By the previous lemma,

$$\text{Chow}_m(\mathcal{F}) \geq \lim_{t \rightarrow \infty} \frac{E(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})}{t}.$$

On the other hand, (X, L) being Chow stable at level m implies that, for some $\varepsilon_0 > 0$ and $C_0 > 0$,

$$E(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}}) \geq \varepsilon_0 (T_m(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})) - C_0.$$

Now using the fact that

$$J_m(\mathcal{F}) = \frac{T_m(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})}{t} \text{ for any } t > 0,$$

we arrive at

$$S(\mathcal{F}) - S_m(\mathcal{F}) \geq \varepsilon_0 J_m(\mathcal{F})$$

for any $\mathbb{N}$ -filtration of R . To extend this inequality to an arbitrary filtration $\mathcal{F}$, for every integer $N \geq 1$ define

$$(\mathcal{F}^{[N]})^\lambda R_k := \mathcal{F}^{\lceil \lambda \rceil / N} R_k.$$

Since $\lceil \lambda_1 \rceil + \lceil \lambda_2 \rceil \geq \lceil \lambda_1 + \lambda_2 \rceil$, this is a multiplicative $\mathbb{N}$ -filtration. Its jumping numbers in degree k are $\lfloor Na_{k,i} \rfloor$. Consequently,

$$0 \leq NS_k(\mathcal{F}) - S_k(\mathcal{F}^{[N]}) < \frac{1}{k},$$

and hence $S(\mathcal{F}^{[N]}) = NS(\mathcal{F})$. For fixed m we also have

$$\frac{S_m(\mathcal{F}^{[N]})}{N} \rightarrow S_m(\mathcal{F}), \quad \frac{J_m(\mathcal{F}^{[N]})}{N} \rightarrow J_m(\mathcal{F}).$$

Apply the already proved inequality to $\mathcal{F}^{[N]}$, divide by N , and let $N \rightarrow \infty$. This yields

$$S(\mathcal{F}) - S_m(\mathcal{F}) \geq \varepsilon_0 J_m(\mathcal{F})$$

for every filtration $\mathcal{F}$, including those with $J_m(\mathcal{F}) = 0$. □

The same rounding construction gives the following extension of Lemma 2.4. The estimate in the proof controls the error linearly in time, which is necessary for passing to slopes at infinity.

Lemma 2.6. *For any filtration $\mathcal{F}$ of R , one has*

$$\text{Chow}_m(\mathcal{F}) \geq \lim_{t \rightarrow \infty} \frac{E(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})}{t}.$$

Proof. Use the filtrations $\mathcal{F}^{[N]}$ introduced above and choose the same compatible orthonormal basis of R_m . Put

$$\psi_{N,t} := \phi_{m,t/N}^{\mathcal{F}^{[N]}}.$$

The inequality $0 \leq a_{m,i} - \lfloor Na_{m,i} \rfloor / N < 1/N$ implies

$$\phi_{m,t}^{\mathcal{F}} - \frac{t}{mN} \leq \psi_{N,t} \leq \phi_{m,t}^{\mathcal{F}}.$$

Thus

$$\|\psi_{N,t} - \phi_{m,t}^{\mathcal{F}}\|_{L^\infty(X)} \leq \frac{t}{mN}, \quad |E_m(\psi_{N,t}) - E_m(\phi_{m,t}^{\mathcal{F}})| \leq \frac{t}{mN}.$$

The supremum-norm Lipschitz property of E gives

$$\left| \frac{E(\psi_{N,t}) - E_m(\psi_{N,t})}{t} - \frac{E(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})}{t} \right| \leq \frac{2}{mN}.$$

The limits as $t \rightarrow \infty$ exist because E is convex along these Bergman subgeodesics and has at most linear growth, whereas E_m is linear in t . Lemma 2.4, with time reparameterized by t/N , gives

$$\frac{\text{Chow}_m(\mathcal{F}^{[N]})}{N} \geq \lim_{t \rightarrow \infty} \frac{E(\psi_{N,t}) - E_m(\psi_{N,t})}{t}.$$

Finally let $N \rightarrow \infty$, using $S(\mathcal{F}^{[N]}) = NS(\mathcal{F})$ and the preceding estimates. □

Corollary 2.7. *For any filtration $\mathcal{F}$ of R with $J_m(\mathcal{F}) > 0$, one has*

$$\frac{S(\mathcal{F}) - S_m(\mathcal{F})}{J_m(\mathcal{F})} \geq \lim_{t \rightarrow \infty} \frac{E(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})}{T_m(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})}.$$

Proof. We conclude using the identity $T_m(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}}) = J_m(\mathcal{F})t$. $\square$

3. FURTHER DISCUSSIONS

In this part we make some general discussions and propose several questions/conjectures for future research.

3.1. Limiting behavior. For any filtration $\mathcal{F}$ of R , it is an interesting question to see if the limit

$$\lim_{m \rightarrow \infty} 2m \cdot \text{Chow}_m(\mathcal{F}) = \lim_{m \rightarrow \infty} 2m(S(\mathcal{F}) - S_m(\mathcal{F}))$$

exists or not. The answer seems to be negative as it turns out to be related to the problem of counting lattice points in certain filtrated Okounkov body (a problem of this sort is already rather non-trivial even for rational polytopes; see e.g. [17]). When $\mathcal{F}$ is a finitely generated $\mathbb{N}$ -filtration of R , the following result is easy to show.

Lemma 3.1. *When $\mathcal{F}$ is a finitely generated $\mathbb{N}$ -filtration of R , the sequence*

$$2m(S(\mathcal{F}) - S_m(\mathcal{F}))$$

has finitely many subsequential limits, indexed by the residue class of m modulo some positive integer. While for any sufficiently divisible $r \in \mathbb{N}_{>0}$, one has

$$\text{Fut}(\mathcal{X}, \mathcal{L}) = \lim_{m \rightarrow \infty} 2mr(S(\mathcal{F}) - S_{mr}(\mathcal{F})),$$

where $(\mathcal{X}, \mathcal{L})$ is some test configuration of (X, L) induced by $\mathcal{F}$.

Proof. Since $\mathcal{F}$ is finitely generated, it implies that the bi-graded ring

$$\bigoplus_{m \in \mathbb{N}} \left(\bigoplus_{\lambda \in \mathbb{N}} \mathcal{F}^\lambda R_m \right)$$

is finitely generated over $\mathbb{C}$. It then follows from the theory of Hilbert functions that

$$\sum_{\lambda \in \mathbb{N}} \dim \mathcal{F}^\lambda R_m = Q(m)$$

for some *quasi-polynomial* Q whenever m is large enough. This means that Q is a polynomial in m , whose coefficients depend on m periodically. On the other hand, by Riemann–Roch we also have for $m \gg 1$

$$d_m = P(m)$$

for some polynomial of degree n . As $\mathcal{F}$ is assumed to be linearly bounded, Q has degree at most $n + 1$. So we can write

$$Q(m) = b_{n+1}(m)m^{n+1} + b_n(m)m^n + \dots + b_0(m),$$

where each $b_i : \mathbb{N} \rightarrow \mathbb{Q}$ is a periodic function. Now using the fact

$$S_m(\mathcal{F}) = \frac{Q(m) - d_m}{md_m} = \frac{Q(m) - P(m)}{mP(m)},$$

we see that the leading coefficient b_{n+1} of Q is a constant depending on $S(\mathcal{F})$ and $\text{vol}(L)$. However starting from the second leading coefficient, each $b_i(m)$ can possibly take finitely many different values. It follows that $2m(S(\mathcal{F}) - S_m(\mathcal{F}))$ converges along each residue class modulo a common period of the coefficients, and therefore has finitely many subsequential limits.

But for any sufficiently divisible $r \in \mathbb{N}_{>0}$, the bi-graded ring

$$\bigoplus_{m \in \mathbb{N}} \left(\bigoplus_{\lambda \in \mathbb{N}} \mathcal{F}^\lambda R_{mr} \right)$$

is generated in degree 1. Then the Hilbert function $Q(mr)$ becomes a polynomial in m whenever m is sufficiently large. In this case one easily sees that $\lim_{m \rightarrow \infty} 2mr(S(\mathcal{F}) - S_{mr}(\mathcal{F}))$ exists and coincides with the generalized Futaki invariant of any test configuration associated to $\mathcal{F}$. $\square$

Corollary 3.2. *Let $\mathcal{F}$ be a finitely generated $\mathbb{N}$ -filtration of R . Then for any sufficiently divisible $r \in \mathbb{N}_{>0}$, one has*

$$\lim_{t \rightarrow \infty} \frac{K(\phi_t^{\mathcal{F}})}{t} \leq \lim_{m \rightarrow \infty} \left(\lim_{t \rightarrow \infty} \frac{K_{mr}(\phi_{mr,t}^{\mathcal{F}})}{t} \right),$$

where

$$K(\phi) := \frac{1}{V} \int_X \log \frac{\omega_\phi^n}{\omega^n} \omega_\phi^n + n \frac{(-K_X) \cdot L^{n-1}}{L^n} E(\phi) - \frac{1}{V} \int_X \phi \operatorname{Ric}(\omega) \wedge \sum_{i=0}^{n-1} \omega^i \wedge \omega_\phi^{n-1-i}$$

denotes the K -energy, extended to the ray by its usual finite-energy extension, and $K_m := 2m(E - E_m)$ is the quantized K -energy. Equality holds if the test configuration induced by $\mathcal{F}$ has reduced central fiber.

Proof. Fix $r_0 \in \mathbb{N}_{>0}$ such that for any multiple r of r_0 , the $\mathbb{C}[t]$ -algebra

$$\bigoplus_{m \in \mathbb{N}} \left(\bigoplus_{\lambda \in \mathbb{Z}} t^{-\lambda} \mathcal{F}^\lambda R_{mr} \right)$$

is generated in degree 1. Then as explained in Remark 2.5, one has

$$2mr(S(\mathcal{F}) - S_{mr}(\mathcal{F})) = \lim_{t \rightarrow \infty} \frac{K_{mr}(\phi_{mr,t}^{\mathcal{F}})}{t} \text{ for all } m \in \mathbb{N}_{>0}.$$

Let $(\mathcal{X}, \mathcal{L})$ be the test configuration induced by $\mathcal{F}$ with exponent r_0 . Applying the previous lemma, for any r divisible by r_0 , we deduce that

$$Fut(\mathcal{X}, \mathcal{L}) = \lim_{m \rightarrow \infty} \left(\lim_{t \rightarrow \infty} \frac{K_{mr}(\phi_{mr,t}^{\mathcal{F}})}{t} \right).$$

Let $\nu : \mathcal{X}^\nu \rightarrow \mathcal{X}$ be the normalization and set $\mathcal{L}^\nu := \nu^* \mathcal{L}$. Normalization does not change the associated geodesic ray, and

$$Fut(\mathcal{X}^\nu, \mathcal{L}^\nu) \leq Fut(\mathcal{X}, \mathcal{L})$$

by [5, Proposition 3.15]. For the normal test configuration, Li's slope formula [15, Theorem 1.7.2], together with the nonnegative central-fiber correction between the non-Archimedean Mabuchi energy and the generalized Futaki invariant, gives

$$\lim_{t \rightarrow \infty} \frac{K(\phi_t^{\mathcal{F}})}{t} \leq Fut(\mathcal{X}^\nu, \mathcal{L}^\nu) \leq Fut(\mathcal{X}, \mathcal{L}).$$

If $\mathcal{X}_0$ is reduced, the total space is regular at the generic points of its central-fiber components: the corresponding one-dimensional local rings have maximal ideal generated by the base parameter, and hence are discrete valuation rings. Thus normalization is an isomorphism in codimension one, its central fiber remains reduced, and both inequalities above are equalities. This proves the assertion without a normality assumption on the original total space. $\square$

When $\mathcal{F}$ is a finitely generated $\mathbb{N}$ -filtration, it follows from Lemma 3.1 that

$$(3.1) \quad Fut(\mathcal{X}, \mathcal{L}) \geq \liminf_{m \rightarrow \infty} 2m(S(\mathcal{F}) - S_m(\mathcal{F})),$$

where $(\mathcal{X}, \mathcal{L})$ is the test configuration of (X, L) induced by $\mathcal{F}$. Equality holds when the central fiber $\mathcal{X}_0$ is reduced: the all-degree lower bound by the non-Archimedean Mabuchi energy established in the proof of Proposition 3.6 below gives the reverse inequality, since in this case that energy equals $Fut(\mathcal{X}, \mathcal{L})$.

For a general filtration $\mathcal{F}$ of R , Lemma 2.6 implies that

$$(3.2) \quad \liminf_{m \rightarrow \infty} 2m(S(\mathcal{F}) - S_m(\mathcal{F})) \geq \liminf_{m \rightarrow \infty} \left(\lim_{t \rightarrow \infty} \frac{2m(E(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}}))}{t} \right).$$

For an $\mathbb{N}$ -filtration, the right hand side is

$$\liminf_{m \rightarrow \infty} 2m(S(\mathcal{F}_m) - S_m(\mathcal{F})),$$

where $\mathcal{F}_m$ is generated by the restriction to R_m , as above. This is twice the invariant $Chow_\infty(\mathcal{F})$ in [25]; the factor 2 comes from our normalization of the quantized K -energy.

Conjecture 3.3. *One has*

- (1) $\liminf_{m \rightarrow \infty} 2m(S(\mathcal{F}) - S_m(\mathcal{F}))$ is equal to the non-Archimedean Mabuchi energy in [5, §7.3];
- (2) $\liminf_{m \rightarrow \infty} \left(\lim_{t \rightarrow \infty} \frac{2m(E(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}}))}{t} \right)$ is equal to the slope at infinity of K along $\phi_t^{\mathcal{F}}$.

If this conjecture holds then combining (3.2) with [15, Theorem 1.7.1] one can deduce that [15, Conjecture 1.6] holds for those maximal geodesic rays induced by filtrations. Probably this conjecture is too good to be true, but at least it is promising for those filtrations that are induced by *models* of (X, L) (in the sense of [15]), in which case we actually expect that $\liminf_{m \rightarrow \infty} 2m(S(\mathcal{F}) - S_m(\mathcal{F}))$ is given by the intersection formula in [16, Theorem 1.1].

Remark 3.4. Given a filtration $\mathcal{F}$ of R , a natural idea is to approximate $\mathcal{F}$ by a sequence of finitely generated filtrations (as in [25]). Then to derive useful information, one key point is to show that the Hilbert function associated to these filtrations stay in a “bounded family”. This seems to be related to the *Fujita approximation conjecture* of Li [16].

3.2. Some new stability thresholds. The following estimate is building on [2, Lemma 7.7] and [24, Lemma 5.2].

Proposition 3.5. *There exists $C_0 > 0$ such that for any filtration $\mathcal{F}$ of R with $J(\mathcal{F}) > 0$, one has*

$$\liminf_{m \rightarrow \infty} \frac{2m(S(\mathcal{F}) - S_m(\mathcal{F}))}{J_m(\mathcal{F})} > -C_0.$$

Proof. Given any $m \gg 1$, we consider the Bergman geodesic ray $\phi_{m,t}^{\mathcal{F}}$. By definition, there exists an H_m -orthonormal basis $\{s_i\}_{1 \leq i \leq d_m}$ of R_m such that

$$\phi_{m,t}^{\mathcal{F}} = \frac{1}{m} \log \sum_{i=1}^{d_m} e^{a_{m,i}t} |s_i|_{h^m}^2,$$

where $a_{m,i}$'s are the jumping numbers of $\mathcal{F}$. Then using the convexity of E along Bergman geodesics, one has for any $t > 0$

$$\begin{aligned} \frac{E(\phi_{m,t}^{\mathcal{F}}) - E(\phi_{m,0}^{\mathcal{F}})}{t} &\geq \frac{d}{dt} \Big|_{t=0} E(\phi_{m,t}^{\mathcal{F}}) = \frac{1}{mV} \int_X \frac{\sum_{i=1}^{d_m} a_{m,i} |s_i|_{h^m}^2}{\sum_{i=1}^{d_m} |s_i|_{h^m}^2} \omega_{\phi_{m,0}^{\mathcal{F}}}^n \\ &= \frac{1}{mV} \int_X \frac{\sum_{i=1}^{d_m} (a_{m,i} - a_{m,d_m}) |s_i|_{h^m}^2}{\sum_{i=1}^{d_m} |s_i|_{h^m}^2} \omega_{\phi_{m,0}^{\mathcal{F}}}^n + \frac{a_{m,d_m}}{m}. \end{aligned}$$

Observe that $\phi_{m,0}^{\mathcal{F}} = \frac{1}{m} \log \sum_{i=1}^{d_m} |s_i|_{h^m}^2$. So by the standard Bergman kernel expansion, one can find $C_1 > 0$ (independent of $\mathcal{F}$) such that

$$\frac{d_m \omega_{\phi_{m,0}^{\mathcal{F}}}^n}{V \cdot \sum_{i=1}^{d_m} |s_i|_{h^m}^2} \leq (1 + \frac{C_1}{m}) \omega^n$$

whenever m is sufficiently large. This implies that

$$\begin{aligned} \frac{E(\phi_{m,t}^{\mathcal{F}}) - E(\phi_{m,0}^{\mathcal{F}})}{t} &\geq \frac{1 + C_1/m}{md_m} \int_X \sum_{i=1}^{d_m} (a_{m,i} - a_{m,d_m}) |s_i|_{h^m}^2 \omega^n + \frac{a_{m,d_m}}{m} \\ &= \frac{1 + C_1/m}{md_m} \sum_{i=1}^{d_m} a_{m,i} - \frac{C_1}{m} \cdot \frac{a_{m,d_m}}{m} \\ &= \frac{1 + C_1/m}{t} E_m(\phi_{m,t}^{\mathcal{F}}) - \frac{C_1}{m} \cdot \frac{T_m(\phi_{m,t}^{\mathcal{F}})}{t}. \end{aligned}$$

Rearranging, we obtain that

$$\frac{E(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})}{T_m(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})} \geq -\frac{C_1}{m} + \frac{E(\phi_{m,0}^{\mathcal{F}})}{T_m(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})} \text{ for any } t > 0.$$

Now letting $t \rightarrow \infty$, we arrive at

$$\lim_{t \rightarrow \infty} \frac{E(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})}{T_m(\phi_{m,t}^{\mathcal{F}}) - E_m(\phi_{m,t}^{\mathcal{F}})} \geq -\frac{C_1}{m},$$

which holds for all $m \gg 1$. Then Corollary 2.7 gives a lower bound $-2C_1$ for the $\liminf$ in the statement. Choosing $C_0 > 2C_1$ proves the asserted strict inequality. $\square$

Therefore, for those $\mathcal{F}$ with $J(\mathcal{F}) > 0$, it makes sense to consider the following quantity:

$$\gamma(\mathcal{F}) := \liminf_{m \rightarrow \infty} \frac{2m(S(\mathcal{F}) - S_m(\mathcal{F}))}{J(\mathcal{F})}.$$

Put

$$(3.3) \quad \gamma(L) := \inf_{\mathcal{F}} \gamma(\mathcal{F}),$$

where $\mathcal{F}$ runs through all the filtrations with $J(\mathcal{F}) > 0$. Note that by Proposition 3.5,

$$\gamma(L) > -\infty.$$

Similarly one can define a quantized version of γ by putting

$$\gamma_m(L) := \inf_{\mathcal{F}} \frac{2m(S(\mathcal{F}) - S_m(\mathcal{F}))}{J_m(\mathcal{F})}.$$

where $\mathcal{F}$ satisfies $J_m(\mathcal{F}) > 0$. By the proof of Theorem 1.8, in the above inf, it is enough to only consider $\mathbb{N}$ -filtrations.

The following proposition relates these thresholds to the usual stability notions. Its second assertion requires an all-degree comparison, because Lemma 3.1 alone only identifies the limit along sufficiently divisible degrees.

Proposition 3.6. *One has*

- (1) (X, L) is Chow stable at level m if and only if $\gamma_m(L) > 0$;
- (2) (X, L) is K -stable if and only if $\gamma(\mathcal{F}) > 0$ for any finitely generated $\mathbb{N}$ -filtration $\mathcal{F}$ with $J(\mathcal{F}) > 0$;
- (3) (X, L) is uniformly K -stable if $\gamma(L) > 0$;
- (4) (X, L) is uniformly K -stable if $\liminf \gamma_m(L) > 0$.

Proof. The first assertion is Theorem 1.8. For the second, we establish the comparison

$$(3.4) \quad M^{NA}(\mathcal{F}) \leq \liminf_{m \rightarrow \infty} 2m(S(\mathcal{F}) - S_m(\mathcal{F})) \leq Fut(\mathcal{X}, \mathcal{L})$$

for every finitely generated $\mathbb{N}$ -filtration, where $M^{NA}(\mathcal{F})$ is the non-Archimedean Mabuchi energy of its associated non-Archimedean metric. The second inequality follows by restricting to sufficiently divisible degrees. We give the details of the first inequality to account for the possible periodic terms in Lemma 3.1.

Let $(\mathcal{Z}, \mathcal{A})$ be the normal ample representative of the metric, with the $\mathbb{Q}$ -polarization $\mathcal{A}$ restricting to L rather than to an exponent multiple of L . On an equivariant smooth common model $p : \mathcal{Y} \rightarrow X \times \mathbb{C}$ and $\rho : \mathcal{Y} \rightarrow \mathcal{Z}$, choose a rational section σ of L and write the corresponding $\mathbb{Q}$ -divisor as

$$\rho^* \mathcal{A} = p^* \operatorname{div}(\sigma) + \sum_E c_E E, \quad b_E := \operatorname{ord}_E(t),$$

where the sum is over the vertical prime divisors. Choose a sufficiently divisible integer q such that $b_E \mid q$ and $qc_E/b_E \in \mathbb{Z}$ for every E . Under the normalized base change $t = \tau^q$, the ramification index along a prime divisor over E is q/b_E . Thus the pulled-back divisor on the normalized base change of $\mathcal{Y}$ is an integral Weil divisor. Pushing it forward to the normalized ample base change $\mathcal{Z}_q$ shows that its $\mathbb{Q}$ -polarization $\mathcal{A}_q$ can be represented by an integral $\mathbb{Q}$ -Cartier Weil divisor. We may also choose q so that $(\mathcal{Z}_q)_0$ is reduced; see [5, Proposition 7.16]. The canonical compactification $\bar{\mathcal{A}}_q$ is still integral Weil, since near infinity it is the actual line bundle L .

Define a filtration $\mathcal{G}$ using the extension lattices

$$H^0(\mathcal{Z}_q, \mathcal{O}_{\mathcal{Z}_q}(m\mathcal{A}_q)) \subset R_m \otimes \mathbb{C}[\tau, \tau^{-1}].$$

Here $\mathcal{O}(m\mathcal{A}_q)$ is the rank-one reflexive sheaf associated to the integral Weil divisor. Each lattice is a free $\mathbb{C}[\tau]$ -module and is equivariant, so its exponents define integer jumping numbers. Choose $r > 0$ such that $r\mathcal{A}_q$ is relatively ample and Cartier. The section modules of the finitely many sheaves $\mathcal{O}(j\mathcal{A}_q)$, $0 \leq j < r$, are finitely generated over the r -th Veronese section algebra, by relative Serre vanishing. Consequently $\mathcal{G}$ is finitely generated.

In every degree we have $q\mathcal{F} \subseteq \mathcal{G}$, where $q\mathcal{F}$ denotes scaling of the weights by q . Indeed, if $s \in \mathcal{F}^a R_m$ with $a \in \mathbb{N}$, a sufficiently divisible power of $t^{-a}s$ is a section of the corresponding Cartier multiple of the original polarization. After pullback, the same holds for a power of $\tau^{-qa}s$. Normality and the integrality of $\mathcal{A}_q$ then imply that $\tau^{-qa}s$ itself is a section of $\mathcal{O}(m\mathcal{A}_q)$: the required inequalities hold at every prime

divisor after dividing the inequalities for that power. In particular, $\mathcal{G}$ satisfies our nonnegative convention, and

$$S_m(\mathcal{G}) \geq qS_m(\mathcal{F}), \quad S(\mathcal{G}) = qS(\mathcal{F}), \quad J(\mathcal{G}) = qJ(\mathcal{F}).$$

The last two equalities follow from invariance of the limiting weight measure under normalization and its scaling under base change; see [5, Theorem 3.14 and Lemma 6.9].

We next verify that $2m(S(\mathcal{G}) - S_m(\mathcal{G}))$ has a limit along all degrees. On $\bar{\mathcal{Z}}_q$, put $A := r\bar{\mathcal{A}}_q$. For $m = kr + j$, $0 \leq j < r$, we have

$$\mathcal{O}(m\bar{\mathcal{A}}_q) = \mathcal{O}(j\bar{\mathcal{A}}_q) \otimes \mathcal{O}(kA).$$

Asymptotic Riemann–Roch for the line bundle A on this normal $(n+1)$ -dimensional variety gives

$$\chi(\mathcal{O}(kA)) = \alpha k^{n+1} + \beta k^n + O(k^{n-1}), \quad \alpha = \frac{A^{n+1}}{(n+1)!}.$$

For the fixed rank-one reflexive sheaf $\mathcal{O}(j\bar{\mathcal{A}}_q)$, its codimension-one class is $j\bar{\mathcal{A}}_q$, so the corresponding expansion is

$$\chi(\mathcal{O}(j\bar{\mathcal{A}}_q) \otimes \mathcal{O}(kA)) = \alpha k^{n+1} + \left(\beta + \frac{j\bar{\mathcal{A}}_q \cdot A^n}{n!} \right) k^n + O(k^{n-1}).$$

For completeness, these first two terms can be computed on a resolution as in [5, Appendix A]: replace the reflexive sheaf by a line bundle agreeing with it off a set of codimension at least two. The difference of its divisor from the pullback of $j\bar{\mathcal{A}}_q$ is exceptional and has zero intersection with the n -th power of the pullback of A . All pushforward and higher-direct-image corrections are supported in codimension at least two and contribute only $O(k^{n-1})$. No $\mathbb{Q}$ -Gorenstein assumption is needed. Substitution of $k = (m-j)/r$ cancels the j -dependent coefficient of m^n . Hence the first two coefficients of $\chi(\mathcal{O}(m\bar{\mathcal{A}}_q))$ are independent of the residue class of m .

Relative Serre vanishing for the same finitely many residue sheaves gives, for all sufficiently large m , a vector bundle

$$\mathcal{V}_m := \pi_* \mathcal{O}(m\bar{\mathcal{A}}_q)$$

on $\mathbb{P}^1$, of rank d_m , with vanishing higher direct images. Under canonical gluing at infinity its degree is the total filtration weight $W_m := \sum_i a_{m,i}(\mathcal{G})$. Therefore

$$W_m = \chi(\bar{\mathcal{Z}}_q, \mathcal{O}(m\bar{\mathcal{A}}_q)) - d_m = b_0 m^{n+1} + b_1 m^n + O(m^{n-1}),$$

with b_0, b_1 independent of m modulo r . The all-degree limit thus agrees with the divisible-degree limit defining the generalized Futaki invariant. Since $(\mathcal{Z}_q)_0$ is reduced, [5, Propositions 7.14–7.16] gives

$$\lim_{m \rightarrow \infty} 2m(S(\mathcal{G}) - S_m(\mathcal{G})) = \text{Fut}(\mathcal{Z}_q, r\mathcal{A}_q) = qM^{NA}(\mathcal{F}),$$

where the Futaki invariant is computed for (X, L) with exponent r . Combining this with $S_m(\mathcal{G}) \geq qS_m(\mathcal{F})$ proves the first inequality in (3.4).

If (X, L) is K-stable and $J(\mathcal{F}) > 0$, the reduced normalized base change has positive Futaki invariant and $J(\mathcal{G}) = qJ(\mathcal{F}) > 0$. Thus $M^{NA}(\mathcal{F}) > 0$ and (3.4) gives $\gamma(\mathcal{F}) > 0$. Conversely, positivity of $\gamma(\mathcal{F})$ for every such filtration gives $\text{Fut}(\mathcal{X}, \mathcal{L}) > 0$ whenever $J(\mathcal{X}, \mathcal{L}) > 0$. When $J = 0$, the normalization is trivial and normalization can only decrease the Futaki invariant, so the original invariant is nonnegative. This proves the second assertion.

For the third assertion, the upper bound in (3.4) gives

$$\text{Fut}(\mathcal{X}, \mathcal{L}) \geq \gamma(\mathcal{F}_{(\mathcal{X}, \mathcal{L})})J(\mathcal{X}, \mathcal{L}) \geq \gamma(L)J(\mathcal{X}, \mathcal{L}).$$

Finally, if $\liminf_{m \rightarrow \infty} \gamma_m(L) > 0$, choose $\varepsilon > 0$ such that $\gamma_m(L) \geq \varepsilon$ for all large m . For a fixed test configuration with $J > 0$, its J_{mr} is positive for large m , and

$$2mr(S(\mathcal{F}) - S_{mr}(\mathcal{F})) \geq \varepsilon J_{mr}(\mathcal{F}).$$

Taking the divisible-degree limit gives $\text{Fut}(\mathcal{X}, \mathcal{L}) \geq \varepsilon J(\mathcal{X}, \mathcal{L})$. The case $J = 0$ follows as above. This proves the fourth assertion. $\square$

When $\mathcal{F} = \mathcal{F}_v$ is induced by a nontrivial valuation $v \in \text{Val}_X$ whose filtration is linearly bounded, one has (cf. [6])

$$\frac{S(\mathcal{F}_v)}{n} \leq J(\mathcal{F}_v) \leq nS(\mathcal{F}_v).$$

For this reason one can also consider

$$\eta(v) := \liminf_{m \rightarrow \infty} \frac{2m(S(\mathcal{F}_v) - S_m(\mathcal{F}_v))}{S(\mathcal{F}_v)}.$$

Remark 3.7. When X is Fano and $L = -K_X$, then for any *dreamy* divisor E over X , one has (cf. [14])

$$\eta(\text{ord}_E) = \frac{A_X(E)}{S(E)} - 1.$$

In this case the $\liminf$ in the definition of η is actually an all-degree limit. To justify this, note that the Rees algebra of the divisorial filtration is normal: it is the intersection of $R[t, t^{-1}]$ with the valuation ring of the corresponding Gauss extension. Moreover, its associated graded algebra is a domain, because orders of vanishing are additive under multiplication. Thus its test configuration has normal total space and an integral reduced central fiber. The initial form of any nonzero section in R_1 is nonzero in that domain. Localizing at this degree-one element shows that $\mathcal{O}(1)$ is a line bundle near the generic point of the central fiber. Away from that fiber it is the actual L . Hence the $\mathbb{Q}$ -polarization is an integral $\mathbb{Q}$ -Cartier Weil divisor, and the residue-sheaf Riemann–Roch argument in the preceding proof applies without base change. The all-degree limit agrees with the divisible-degree limit. Fujita’s formula [14, Theorem 5.2] then gives

$$\lim_{m \rightarrow \infty} 2m(S(E) - S_m(E)) = A_X(E) - S(E),$$

which proves the stated identity for η . The expansion in [13, Proposition 4.1] alone is formulated for sufficiently divisible degrees; the preceding argument accounts for all remaining residue classes. See also [8] for the case of general polarizations.

We end this note by asking the following

Question 3.1. *Under what conditions is the inf in (3.3) computed by some finitely generated $\mathbb{N}$ -filtration?*

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LABORATORY OF MATHEMATICS AND COMPLEX SYSTEMS, SCHOOL OF MATHEMATICAL SCIENCES, BEIJING NORMAL UNIVERSITY, BEIJING, 100875, PEOPLE'S REPUBLIC OF CHINA.

Email address: kwzhang@bnu.edu.cn